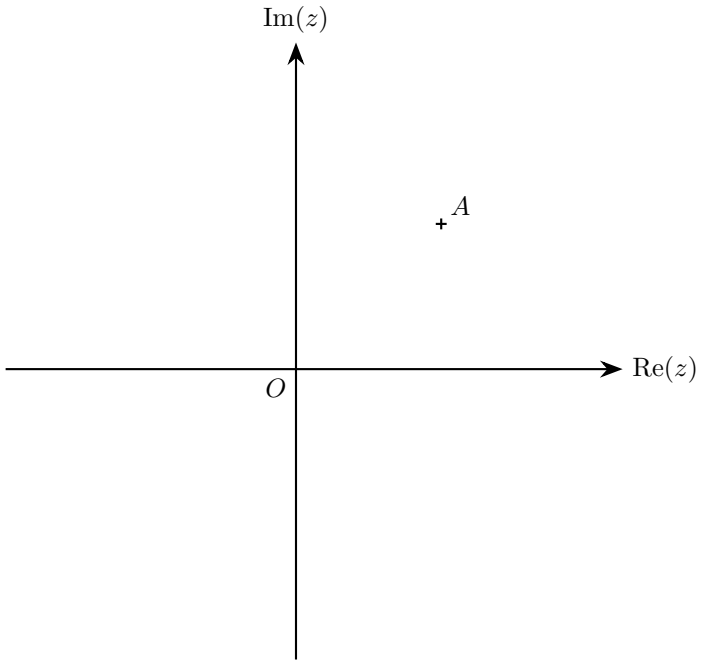
CORE PURE / COMPLEX NUMBERS

Complex Roots and Geometry Problems

Questions by topic 13 questions · 127 marks

Questions

Question 1	*8 marks*	2
Question 2	*11 marks*	3
Question 3	*9 marks*	4
Question 4	*10 marks*	5
Question 5	*5 marks*	6
Question 6	*10 marks*	7
Question 7	*7 marks*	8
Question 8	*12 marks*	9
Question 9	*9 marks*	10
Question 10	*13 marks*	11
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Question 12	*8 marks*	13
Question 13	*11 marks*	14



1. An Argand diagram with the point A representing a complex number z_1 is shown in the diagram. The complex numbers z_2, z_3 and z_4 are $z_1 e^{\frac{1}{2}\pi i}$, $z_1 e^{\pi i}$ and $z_1 e^{\frac{3}{2}\pi i}$ respectively.

(a) (i) On a copy of the Argand diagram, mark the points B, C and D representing the complex numbers z_2, z_3 and z_4. **[2]**

(ii) Show that $z_1 + z_2 + z_3 + z_4 = 0$. **[2]**

(b) Given now that z_1, z_2, z_3 and z_4 are roots of the equation $z^4 = -16$, find these four roots, giving your answers in the form $a + ib$, where a and b are real and exact. **[4]**

2. The five solutions of the equation $z^5 = 16 + 16\sqrt{3}\,i$ are represented by points on an Argand diagram.

These points are the vertices of a regular pentagon P with centre at the origin.

(a) Determine the five complex numbers that represent these vertices, giving your answers in the form $re^{i\theta}$, where $0 \leq \theta < 2\pi$. **[5]**

(b) Sketch P on an Argand Diagram, taking care to label each vertex. **[2]**

(c) Show that the area of P can be written $k \sin \frac{2\pi}{5}$ where k is an integer to be determined. **[4]**

3. In an Argand diagram, the points A and B are represented by the complex numbers $1 + 4i$ and $7 + 2i$ respectively. The circle C passes through A and B, and its centre lies on the real axis.

(a) Find the equation of C, giving your answer in the form

$$|z - a| = b \qquad a \in \mathbb{C},\ b \in \mathbb{R}$$

[4]

The circle D, with equation $|z - (1 + i)| = 5$, intersects C at the points representing the complex numbers z_1 and z_2.

(b) Find the complex numbers z_1 and z_2. **[5]**

4. Given that $z_1 = 7 - i$ and $z_2 = \dfrac{z_1}{1-i}$,

(a) find z_2 in the form $a + bi$, where a and b are real. **[2]**

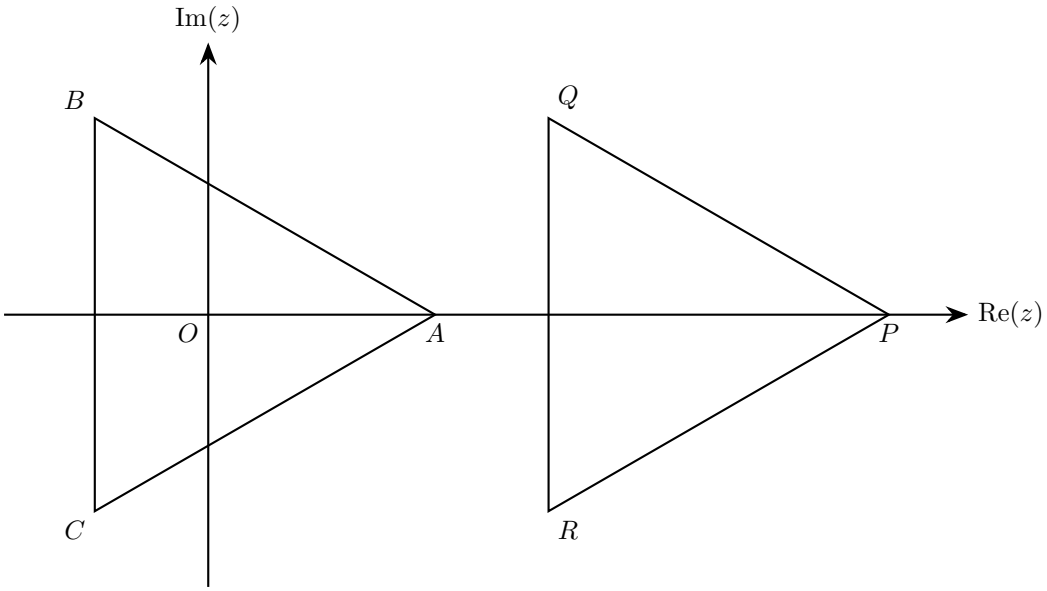
(b) Show on an Argand diagram the point P representing z_1 and the point Q representing z_2. **[2]**

(c) Given that O is the origin, show that $\angle OQP = \dfrac{\pi}{2}$. **[2]**

The circle passing through the points O, P and Q has centre C. Find

(d) the complex number represented by C. **[2]**

(e) the exact value of the radius of the circle. **[2]**



5. The cube roots of unity are represented on the Argand diagram above by the points A, B and C.

The complex numbers represented by P, Q and R are obtained by adding 2 to the complex numbers represented by A, B and C respectively.

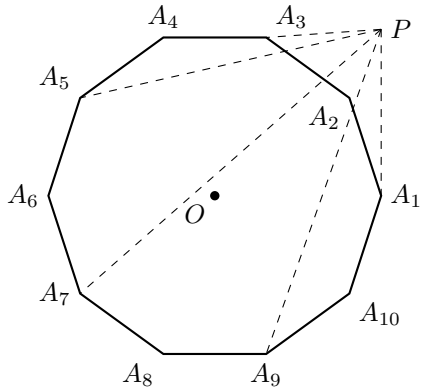
Determine a degree 6 polynomial equation with integer coefficients whose roots are the complex numbers represented by the points A, B, C, P, Q and R. **[5]**

6. In an Argand diagram, the points A, B and C are the vertices of an equilateral triangle with centre G. The point G represents the complex number $1-2i$, and the point A represents the complex number $5+2i$.

(a) Find the complex numbers represented by the points B and C, giving your answers in the form $x + iy$, where x and y are real and exact. **[6]**

The points D, E and F are the centroids of triangles GBC, GCA and GAB respectively.

(b) Find the exact area of triangle DEF. **[4]**



7. **(a)** Using the identity $zz^* = |z|^2$, or otherwise, show that if w is any root of unity then

$$|w - (1+i)|^2 = 3 - (1-i)w - (1+i)w^* \qquad \textbf{[3]}$$

The diagram shows a regular decagon $A_1A_2 \ldots A_{10}$ whose vertices all lie on the unit circle with centre at the origin O and A_1 at $(1, 0)$. The point P lies in the same plane as the decagon and has coordinates $(1, 1)$.

(b) Using the result from part (a), find

$$\sum_{r=1}^{5} (PA_{2r-1})^2 \qquad \textbf{[4]}$$

8. You are given the complex number $\alpha = \cos\frac{3\pi}{5} + i\sin\frac{3\pi}{5}$ and the equation $z^5 = -1$.

(a) Show that α is a root of the equation. **[2]**

(b) Write down the other four roots of the equation. **[1]**

(c) Show that

$$\alpha^4 - \alpha^3 + \alpha^2 - \alpha + 1 = 0$$ **[2]**

(d) Hence show that

$$\left(\alpha + \frac{1}{\alpha}\right)^2 - \left(\alpha + \frac{1}{\alpha}\right) - 1 = 0$$ **[3]**

(e) Hence determine the value of $\cos\frac{3\pi}{5}$ in the form $a + b\sqrt{c}$, where a, b and c are rational numbers to be found. **[4]**

9. The points representing the complex numbers $z_1 = 6 + 11i$ and $z_2 = -8 - 3i$ are opposite vertices of a regular octagon, K, in the complex plane.
The centre of K represents the complex number α.

(a) Show that $\alpha = -1 + 4i$. **[2]**

Given that

$$\beta = \frac{1 - i}{14}$$

(b) Show that

$$\beta(z_1 - \alpha) = 1$$ **[2]**

The vertices of K are given by the roots of the equation

$$(\beta(z - \alpha))^8 = 1$$

(c) (i) Write down the roots of the equation $w^8 = 1$ in the form $re^{i\theta}$. **[1]**

(ii) Hence, or otherwise, determine the positions of the other six vertices of K, giving your answers as complex numbers in Cartesian form. **[4]**

10. (i) Find the 8th roots of $-128 + 128\sqrt{3}\,i$ in the form $re^{i\theta}$, where $r > 0$ and $0 \leq \theta < 2\pi$. **[5]**

(ii) These 8th roots are represented in the Argand diagram, in order of increasing θ, by the points A, B, C, D, E, F, G, H.
Draw the Argand diagram, making clear which point is which. **[2]**

(iii) The mid-point of BD is the point P which represents the complex number w.
Find, in exact form, the modulus and argument of w. **[3]**

(iv) w is an nth root of a real number a, where n is a positive integer. State the least possible value of n and find the corresponding value of a. **[3]**

11. The cubic equation

$$z^3 + pz^2 + qz + 24 = 0$$

where p and q are real, has roots α, β and γ.
It is given that $\alpha = -2 + 2i$.

(a) (i) Write down another root, β, of the equation. **[1]**

(ii) Find the third root, γ. **[3]**

(iii) Find the values of p and q. **[3]**

(b) (i) Express α in the form $re^{i\theta}$, where $r > 0$ and $-\pi < \theta \leq \pi$. **[2]**

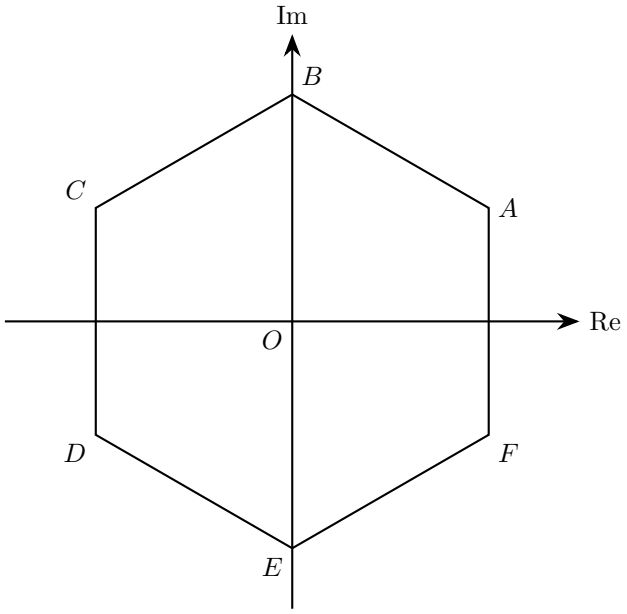
(ii) Hence show that

$$(-2 + 2i)^n = (2\sqrt{2})^n \left(\cos \frac{3n\pi}{4} + i \sin \frac{3n\pi}{4} \right)$$ **[2]**

(iii) Hence show that

$$\alpha^n + \beta^n + \gamma^n = 2(2\sqrt{2})^n \cos \frac{3n\pi}{4} + (-3)^n$$

where n is an integer. **[3]**



12. In the diagram, the points A, B, C, D, E and F represent the complex sixth roots of -729 on an Argand diagram. The centroids of triangles OAB, OBC, OCD, ODE, OEF and OFA are G, H, I, J, K and L respectively.

(a) Write down, in exponential form, the complex numbers represented by the points A, B, C, D, E and F. **[2]**

(b) When these complex numbers are multiplied by the complex number w, the resulting complex numbers are represented by the points G, H, I, J, K and L.
Find w in exponential form. **[4]**

(c) You are given that G, H, I, J, K and L represent roots of the equation $z^6 = p$.
Find p. **[2]**

13. (i) Write down, in any form, all the complex roots of the equation

$$w^{11} = -1$$ **[2]**

(ii) Explain why the equation

$$(z+1)^{11} = -(z-1)^{11}$$

has exactly 10 non-real roots and show that they may be expressed in the form

$$-i\cot\left(\frac{m\pi}{22}\right)$$ **[6]**

where $m = \pm 1, \pm 3, \pm 5, \pm 7, \pm 9$.

(iii) Show that

$$\left\{-i\cot\left(\frac{m\pi}{22}\right)\right\}\left\{i\cot\left(\frac{m\pi}{22}\right)\right\} = \cot^2\left(\frac{m\pi}{22}\right)$$ **[1]**

(iv) Given that the product of the non-zero roots of the equation in part (ii) is 11, find the value of

$$\tan^2\left(\frac{\pi}{22}\right)\tan^2\left(\frac{3\pi}{22}\right)\tan^2\left(\frac{5\pi}{22}\right)\tan^2\left(\frac{7\pi}{22}\right)\tan^2\left(\frac{9\pi}{22}\right)$$ **[2]**